Altera Labs:

Launching a Verifiable AI Tutoring System for STEM Education

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Program Duration	Funding Requested
9 Months	**$49,638 USD**

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Section A. Technology Assessment

Problem Statement: Artificial intelligence in education is facing a pedagogical crisis. While student adoption is mainstream (86% of students use AI)[1], they primarily use "stateless answer machines" that encourage a passive "substitution" model of learning. Recent studies indicate that reliance on AI tools can reduce critical thinking skills (75%), encourage overreliance on technology (73%), and propagate misinformation (70%)[2], with over half of STEM learners admitting to bypassing their own problem-solving processes—an approach that "falls short in fostering… advanced critical thinking."[3] This is compounded by existing non-AI software (like WeBWorK) that provides only binary "right/wrong" feedback, creating a "black box" of learning loss with no data for instructor intervention.

Solution Statement: Altera Labs directly addresses this crisis via a **state-aware AI tutoring system**. Our core innovation is an **Adaptive Knowledge Tracking** engine. Instead of offering stateless answers, our platform constructs a dynamic and personalized model of **each student's knowledge, misconceptions, and learning progression**. This approach fosters active, guided problem-solving, ensuring students achieve a deep understanding of concepts and **equipping faculty with the precise data** needed to proactively intervene and optimize their teaching.

Technology Description and Status: The Altera Labs platform is at **TRL 3 (Validated Components)**. We have successfully de-risked both pedagogy and technology, confirming strong faculty demand via a JHU Honors Analysis demo (see attached letter) while completing our core TRL 3 components **(~15,000 lines of code, 65-endpoint Flask API)**. We are now integrating these components into a TRL 4 prototype for a technical pilot in JHU's Honors Analysis in time for final exam season. The BII grant will fund the critical next step: advancing this TRL 4 pilot to a robust, TRL 5 Alpha to generate the performance and user data required for commercialization. Our solution is implemented through a proprietary, five-phase architecture designed to manage the full learning lifecycle.

Phase 1-2: AI-Driven Content Ingestion and Pedagogical Moat

1. **Content Ingestion (Phase 1):** We use PyMuPDF to extract text and then generate 768-dimensional vector embeddings using Google's Vertex AI.
2. **Quiz & Answer Key Generation (Phase 2):** A multi-agent ControlFlow service analyzes the ingested content to generate personalized quizzes and **complete answer keys**. These keys serve a critical dual purpose: they provide rich, immediate feedback to the **student** (via detailed explanations, 3-level hints, and plausible misconceptions) and simultaneously serve as the "ground truth" for our **AI tutor**, informing its Socratic questioning to ensure pedagogical accuracy.

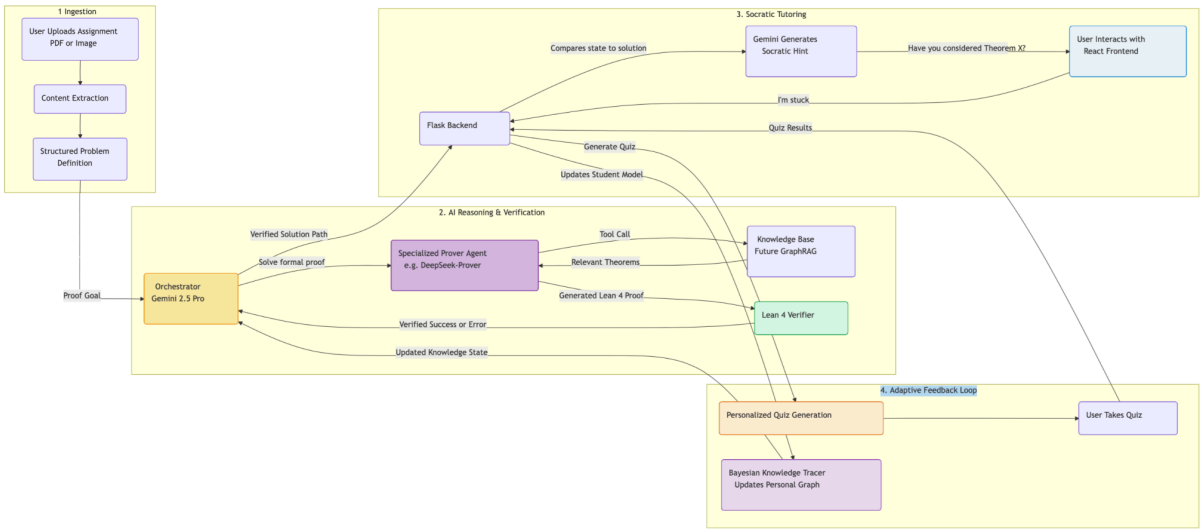

Figure 1: Architectural diagram for our proprietary multi-agent AI system, *showing the flow from ingestion to verification and tutoring.*

Phase 3-4: The "Adaptive Brain" & Real-Time Knowledge Tracing

When a student submits an answer (Phase 3), our system validates it server-side and triggers the Bayesian Knowledge Tracing (BKT) engine (Phase 4). This "state-aware" engine uses a **precision-weighted Bayesian fusion** algorithm to mathematically combine the student's prior knowledge with the evidence from their new answer. The result is a new, statistically robust probability of mastery (μ) and uncertainty (σ) for each concept. This updated state is visualized in real-time on the interactive student and faculty dashboards.

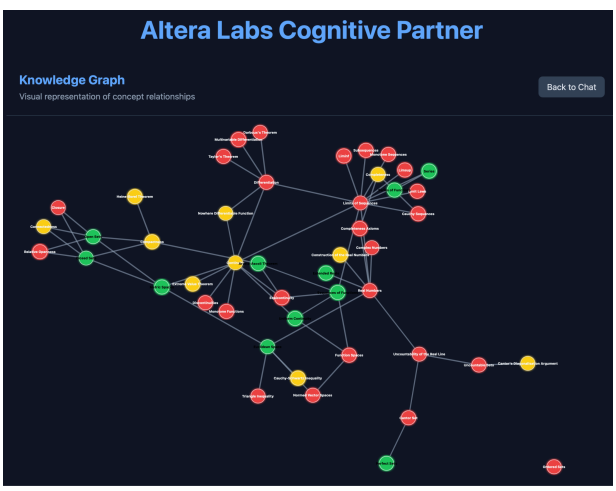

Figure 2: Proprietary BKT 'Course Knowledge Map' *generated for JHU's Honors Analysis class. Circular nodes track mastery (green = mastered, yellow = almost-mastered, red = requires improvement) of core concepts, and edges highlight concept relationships.*

Phase 5: The "Mathematical Judge": Our Verifiable Trust Moat

To build the institutional trust required to scale, our beachhead market is proof-based mathematics. This domain directly addresses the critical trust gap in AI, as research shows students worry about "plausible-sounding yet incorrect responses" and rate their trust in information accuracy from AI 11% lower than from human tutors.[4]

Our "Mathematical Judge" (Phase 5) is a functional **1,841-line module** built to solve this. It is not an AI grader; it is a **non-AI verification engine** that uses **Lean 4**, the objective proof assistant trusted by the formal mathematics community. Inspired by

Google's DeepMind's use of a Lean-based system to achieve a near-gold-medal level at the International Math Olympiad, our system uses mathematical logic to certify the correctness of a proof and eliminate any risk of AI hallucination. Our autonomous 5-strategy solver achieves a **100% success rate (6/6)** on our number theory test suite, constructing and verifying a formal proof in an average of just **15.92 seconds**. This solver is guided by our **'Verifiable Knowledge Library,' a proprietary 559,000-concept knowledge graph powered by GraphRAG** that finds relevant theorems and dependencies to use.

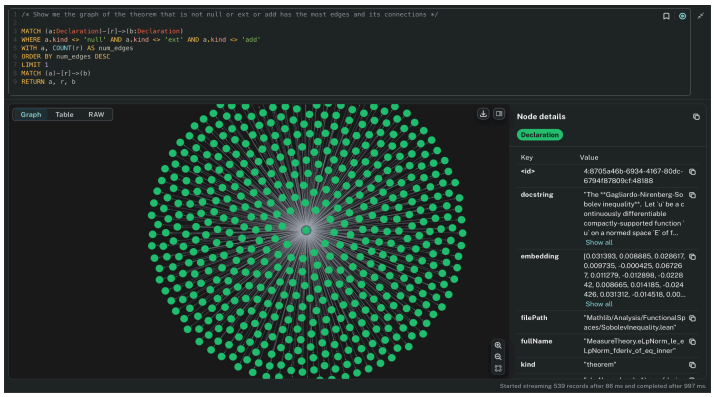

Figure 3: Visualization of our "Verifiable Knowledge Library," *showing the theorem 'Gagliardo-Nirenberg-Sobolev inequality' (center) and all its dependencies in the 559,000-concept library.*

Only after this non-AI verification does a specialized AI use the answer key to guide the student through Socratic Questioning. This process implements the "hybrid educational model" that research has shown to be an optimal pedagogical strategy.[5] Our "state-aware" AI provides the non-judgmental, accessible, and safe space that students value in AI tools, while simultaneously emulating the deep inquiry and critical thinking facilitation that students value in expert human tutors.

This verification-first architecture provides the deep institutional trust required to scale our core Adaptive Brain (BKT) and "intelligent library" (GraphRAG) technology. The BII grant will help enable the creation of new knowledge libraries for other subjects (e.g., Physics). For these non-verifiable subjects, we will replace the "Mathematical Judge" with an AI panel of experts consensus model, providing a **high-accuracy, scalable path to all of STEM.**

Section B. Market Analysis and Competition

Market Size and Trends: Altera Labs targets the U.S. higher-education learning market. Our primary target market includes higher education institutions with STEM disciplines, serving both students and instructors who benefit most from adaptive knowledge tracking and learning analytics. Our secondary market includes other quantitative domains such as economics and data science.

Using a **per-student licensing model**, the **Total Addressable Market (TAM)** is estimated at $554 million with an annual license rate of $30 per seat (18.4 M postsecondary students).[6] The **Serviceable Available Market (SAM)** focuses on 3.3 million STEM students at institutions with AI access, representing a $100 million

opportunity. With an anticipated market penetration rate of 3% representing an estimate of STEM students in Baltimore/Maryland, our initial **Serviceable Obtainable Market (SOM)** targets 100,000 students, a $3 million market capture.[7] Multiple macro-trends reinforce this opportunity. STEM degree awards have surged (master's in S&E fields rose 41% from 2012-2021), expanding the learner base for advanced education.[8] Meanwhile, although most students use AI for study, over 50% rely on it to produce answers and bypass the problem-solving process, which has negative impacts on student learning outcomes.[9] These trends point to a fast-growing demand for AI systems that foster understanding, not shortcuts.

Value Propositions: For students, we offer a **personalized tutor** that enhances conceptual understanding and retention. **Our B2B value proposition** for instructors is validated by Dr. Kobe Marshall-Stevens (JHU Math), our Faculty Champion. He highlights that in large classes, instructors struggle to provide **detailed, individualized feedback**, often outsourcing grading to graduate students with limited teaching experience. This delay in feedback, he confirms, is **"precisely the gap that Altera Labs seeks to address,"** offering an **"indispensable tool" for instructors**. Dr. Marshall-Stevens further validates our scalability, noting that "since the platform requires little to no input from the instructor, I strongly believe that the platform has potential to scale to large-scale classes with hundreds of students."

Competition Analysis: Altera Labs differentiates itself with its **"glass-box" and state-aware architecture**, guiding learners through step-by-step questioning and continuously mapping their knowledge state ("Adaptive Brain") to provide targeted, data-driven support for instructors. Our two-fold technical advantage over competitors includes:

- **Superior Retrieval:** Unlike traditional RAGs used by competitors like StudyFetch, our "Verifiable Knowledge Library" uses GraphRAG. This offers more accurate, context-aware retrieval and multi-hop reasoning.
- **Guaranteed Verification:** For our math beachhead, our "Mathematical Judge" formally verifies proof logic, ensuring near 100% correctness beyond mere information retrieval.

Competitive Analysis Matrix

Capability	Altera Labs	Khanmigo (Khan Academy)	Math, Inc. (Gauss)	StudyFetch	Roadmap AI (Roadmap.sh)
Interactive Tutoring	Strong	Strong	Limited	Strong	Strong
Personalized Knowledge Tracing	Strong	Strong	Limited	Moderate	Moderate
Answer Verification	Strong	Limited	Strong	Moderate	Moderate
Primary Market Segment	Higher education faculty and undergraduate students	K-12 and early undergraduate learners	Graduate-level researchers and research institutions	K-12 and undergraduate students	Independent learners and software developers

Section C. Commercialization Pathway and Risk Assessment

Commercialization Pathway: Our primary commercialization pathway is a startup-led B2B departmental license. Our notional business model is built on a per-student, annual license fee **(preliminarily estimated at $30/student)**, a figure derived from our market analysis of similar products. Our operational value proposition is twofold: **To Deans & Department Chairs (e.g., Dr. Brown)**, we provide the "Course Knowledge Map" Dashboard, an LMS plugin that provides real-time, aggregate analytics to identify at-risk students and improve retention in gateway STEM courses. **To Faculty (e.g., Dr. Marshall-Stevens)**, we provide a pedagogical tool that moves assessment from episodic (midterms, finals) to continuous, revealing *why* students are struggling and saving faculty time.

Our go-to-market strategy is bifurcated. **Phase 1** is the BII-funded technical validation pilot at Johns Hopkins. To secure a departmental Letter of Intent (LOI) from this pilot, we must generate specific efficacy data by hitting key commercial benchmarks, including: (1) Demonstrating a faculty-validated improvement in student concept mastery via the BKT dashboard, and (2) Achieving a >95% success rate for the 'Mathematical Judge' on course homework. **Phase 2**, running in parallel, leverages our team's NSF I-Corps **expertise** to execute a targeted **B2B commercial validation campaign**. This campaign is not for initial discovery, but to **pressure-test our core business model hypotheses** (e.g., the $30/student price point and the value of retention analytics to Deans) with stakeholders at BII partner institutions (e.g., Morgan State, UMBC) and co-design future paid pilots. Key technical and commercial milestones for this 9-month grant include: (1) completing the JHU pilot technical validation, (2) securing a signed LOI from the JHU Math Department, and (3) finalizing a "Joint Pilot Co-Design" document signed by a faculty champion from at least one BII partner institution.

Funding To-Date and Founding Team: As a pre-seed venture, Altera Labs has secured $5,500 of non-dilutive support and training from the JHU Pava Center's competitive Spark and Fuel accelerator programs. The founding team is uniquely poised to execute, led by Peter Seelman (CEO), an AI researcher with experience at JHU Applied Physics Lab (APL) and NASA; Alex Kroumov (CTO), a JHU BME and Applied Math researcher; Akira Lonske (PI/Lead Developer), an ex-Amazon software engineer; and Qiyue (Chris) Yao (Strategy), a Carey Business School Venture Fellow pursuing a MS in Business Analytics & AI. The founders are alumni of the NSF I-Corps program, bringing proven expertise in multi-agent AI systems, AI verification, software development, rigorous B2B customer discovery, and fundraising. They have collectively raised over $190,000 for previous ventures in grants from the State of Arizona, JHU APL, JHU, VentureWell, NIH, and pitch competitions.

Advisory Team: Our advisory board provides unparalleled domain, institutional, and commercial validation. We are actively partnered with **Dr. Emily Riehl** (JHU Math), a leading expert on formal verification (mathlib) and AI-generated math, who provides unmatched insight into competitor weaknesses. **Dr. Richard Brown** (JHU Math Dean of Undergraduate Studies) is our "voice of the customer," guiding our institutional adoption, pricing, and procurement strategy. **Dr. Kobe Marshall-Stevens** (JHU Math) is our primary Faculty Champion, providing direct access to our BII pilot cohort and the "ground truth" data needed to validate our knowledge tracing engine. **Jennifer Harvey** (Ex-Google Accelerator Program Manager), a 15-year Google veteran, guides our brand strategy and community-building to de-risk early adoption.

Risk Assessment: We have actively identified and mitigated key commercialization risks. **Technical Risk** (AI Hallucination) is mitigated by our mathlib GraphRAG, a domain-specific system grounded in a custom-built 559,000+ knowledge graph based on the formal library, not a general-purpose AI. **Market Adoption Risk** (Faculty Rejection) is de-risked by our B2B value prop—the "Course Knowledge Map"—which provides direct operational value to faculty and is championed by the JHU DUS (Dr. Brown). **Institutional Mismatch Risk** (a "One-Size-Fits-All" failure) is addressed by our Phase 2 Co-Design Plan; we are using BII funds for a 9-month structured interview process to **validate** our model with BII partners before deploying, ensuring institutional alignment. **Timeline Risk** (Overly Aggressive Scope) is managed by our de-risked, phased milestones: (1) JHU technical validation and (2) BII-partner co-design, which are two distinct, achievable 9-month goals. **Competitive Risk** (Market Incumbents) is mitigated by our niche-to-scale strategy; our proof-based math beachhead is a high-trust, low-fault-tolerance market that general AI avoids, allowing us to build institutional trust by being the first to collaborate with faculty.

Section D. Project Milestones and Detailed Budget/Justification

Our 9-month project is a "dual-track" sprint for the integration, testing, and validation required for commercialization.

Track 1: Technology Advancement (R&D): The R&D milestones below are explicitly designed to build upon our current, faculty-validated demo and produce the first set of formal, quantifiable performance data (M4) required to secure paid pilots.

(M1-M3) Milestone 1: Integrate BKT Engine & Quiz System.

- **Description:** Functionally connect the BKT engine to the Quiz Service. Validate its accuracy against anonymized data from JHU's Honors Analysis course.
- **Deliverable:** *Technical Report 1:* BKT Integration demonstrating a 100% successful round-trip of quiz data to frontend graph update with >90% accuracy.

(M2-M4) Milestone 2: Integrate 'Mathematical Judge' API.

- **Description:** Integrate the 'Mathematical Judge' module with our frontend by connecting our GraphRAG library to the Lean 4 verifier and AI tutor. Test it against a 100-proof test suite from JHU Calculus & Honors Analysis homework.
- **Deliverable:** *Technical Report 2: 'Mathematical Judge' Validation,* showcasing that the "Mathematical Judge" module achieves (a) >95% verification accuracy on the 100-proof test suite and (b) an average data retrieval time of <5 seconds to the UI (reduced from the current 10-second baseline).

(M4-M6) Milestone 3: Integrate Core Components.

- **Description:** This milestone unifies M1 and M2. We will integrate the full-stack system, connecting all service components into a fully functional, end-to-end Alpha Prototype that may be implemented in a class setting.
- **Deliverable:** A **functional, end-to-end Alpha Prototype** for user testing.

(M4-M9) Milestone 4: Conduct Small-Scale Technical Validation.

- **Description:** Conduct a technical validation study with 50 JHU students from Dr. Marshall-Stevens's Calculus course to gather end-user performance data on the integrated Alpha Prototype.
- **Deliverable:** *Technical Report 3: Alpha Prototype Validation & User Feedback*, demonstrating the following performance metrics in the user study: (a) >95% verification accuracy on student-submitted proofs, and (b) an average total processing time of <10 seconds from student submission to Socratic feedback.

Track 2: Commercial Validation & Admin: In parallel, we will execute our commercialization plan (Section E) to validate the market and co-design future pilots.

- **(M1) Formally register "Altera Labs"** as a Maryland entity.
- **(M1-M6) Conduct 50+ additional B2B validation interviews** with stakeholders (Deans, DUSs, IT Admins, Faculty) at Morgan State, UMBC, and Towson to **pressure-test our core commercial hypotheses** (e.g., $30/student pricing, value of retention analytics).
- **(M7) Complete a "Technical Integration & Procurement Report"** based on university IT and departmental buyer interviews to de-risk future adoption.
- **(M8) Secure 3 "Joint Pilot Co-Design" agreements** or formal Letters of Interest from faculty champions at BII partner institutions, validating demand.

MID-TERM REPORT (4.5 Months): Submit (1) **BKT Engine Validation Results** (M1), (2) **'Mathematical Judge' Module Validation Results** (M2), (3)

Commercial Hypothesis Validation Report (summarizing findings from M1-M6 validation interviews), and (4) LLC Articles of Incorporation Summary.

FINAL REPORT (9 Months): Submit (1) **Final Technical Validation Report** (M4), (2) **A Signed Letter of Intent (LOI)** from the JHU Math Department for a paid 2026-2027 pilot, and (3) **A "Joint Pilot Co-Design" Document** signed by a faculty champion from at least one BII partner institution (e.g., Morgan State).

Section E. Commercialization Plan

This BII-funded plan is designed to **validate the key commercial hypotheses** that emerged from our preliminary (unfunded) discovery and faculty partnerships at JHU:

1. **Economic Buyer:** Deans and Department Chairs (not faculty) are the economic buyers who control departmental budgets for pedagogical tools.
2. **Value Proposition:** The "Course Knowledge Map" and its quantifiable impact on student retention (a key university metric) is the primary purchasing driver for economic buyers.
3. **Pricing:** A **$30/student license** (derived from our TAM/SAM model) is a viable price point for gateway STEM courses.
4. **Technical-Commercial Risk:** Successful adoption requires a low-friction, secure integration path, likely a non-intrusive LMS (Canvas/Blackboard) plugin.

Baltimore Equitable Innovation Pilot & Validation Plan: We will execute our **Track 2 commercial milestones**. We will use the $3,500 travel budget and $3,000 ambassador stipends (see Budget) to conduct targeted validation interviews at Morgan State, UMBC, and Towson. The objective is to **pressure-test our Core Commercial Hypotheses (1-4)** with Deans, DUSs, and (critically) IT administrators to produce our "Technical Integration & Procurement Report" (M7).

- **Deliverable 1:** A "Joint Pilot Co-Design" Document (M8) signed by a faculty champion from at least one BII partner institution, validating institutional demand and alignment.
- **Phase 2: JHU Technical & Pedagogical Validation (Months 4-9):** We will work with our confirmed JHU faculty champions (Dr. Marshall-Stevens, Dr. Brown, Prof. Riehl) to deploy the Alpha Prototype with another JHU pilot cohort.
- **Deliverable 2:** Generate the specific efficacy data from Milestone 4 (M4) required to secure a **signed Letter of Intent (LOI) from the JHU Math Department** for a paid 2026-2027 contract.

Company Formation: Within Month 1 of the grant period (M1), we will allocate funds from the "Legal Fees" budget to file the Articles of Organization with the Maryland SDAT, formally registering Altera Labs as a Maryland-based company.